

AAROHAN 2026

Ascent of Knowledge

XII

Sample
Paper

Name

Roll No.

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Instructions: -

- Read the passage carefully, it contains all the necessary details for solving the questions that follow it.
- The questions will be based on Physics, Chemistry, Mathematics and Analytical Skills.
- The questions can test your understanding of one or more of the above subjects at once.
- Questions marked with an asterisk(*) **may** have more than one option as the correct answer(s). You must select all correct option(s) to earn marks.
- **All other questions have only one correct alternative.** The marking scheme is
 - +3 -> Correct Answer
 - 0-> Unattempted Question
 - -1 -> Incorrect Answer
- For **multiple correct answer** type questions the marking is scheme is
 - +4 -> All correct options selected
 - 0 -> All other cases
- If an assumption is to be made, it will be mentioned in the question itself, along with all other necessary data/information required. Try to crack with the information that has been given- we test your **thinking process** not rote knowledge.
- Before you start this exam, **calm down** and remind yourself that **you can crack this! Keep moving forward, do not get stuck on one question.**
- **Best of Luck!!**



INTRODUCTION

The Human Body is the most sophisticated machine on this planet. Our own body continues to fascinate us as much as the vastness and strangeness of outer space. The intricate arrangement and interconnected functioning of our internal organ systems is, to say the least, extremely intriguing. Sorry for the silly (and obvious) alliteration here, but you get the point. So, studying the biology behind our body is, well, best left to the biologists. Let us start with looking at the P, C and M of the simplest functions like walking, running, etc., then moving on to important organ systems and their functions, and then to Sports, and how we apply our knowledge of Science in analyzing it.

PART 1

This part comprises two passages and questions. Right so, let us begin with a physical model of walking and running. Believe me, creating “models” is how scientists deal with all of their scientific problems. Our simplistic model makes the following assumptions: -

1) The legs are stiff and oscillate at their natural resonant frequency. Newton’s second law is solved assuming the mass (m) of the body is concentrated at the top of the legs (length L) and moves forward with a constant horizontal velocity (v_x), called the walking speed/running speed. The transition from walking to running occurs when both feet lose contact with the ground. If the legs are stiff during walking, the body moves forward through a sequence of circular arcs, like a wheel rotating on its spokes without benefit of rim. Each foot is in contact with the ground 50% of the time and leaves the ground at the instant the other touches. As a result, the centre of mass of the body, assumed concentrated at the top of the legs, moves forward along a trajectory as shown below (Image 1):

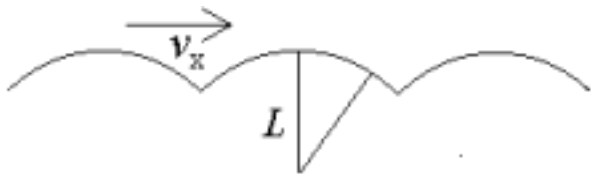


Image1: Conjoined Circular Arcs of Centre of Mass while walking, these assume parabolic shape with maximum range while running.

Now, during the circular trajectory of the centre of mass, it has a net, approximate acceleration v_x^2/L downwards. So, the normal force N acting on it is

$$N = mg - mv_x^2/L, \text{ since}$$

weight and N are the only two vertical forces. From here, we can see that there are speeds v_x for which N becomes zero and weight is the only force. This is our transition from



walking to running. The speed at which this occurs is called Transition speed.

During walking or running, let us turn our attention to the sharp cusps where the motion suddenly changes direction (downward to upward). This part of the motion is propelled by an impulse J equal to the integral of the downward force over the remaining interval, so that the net average acceleration is zero. Hence $J = (mg - N)T/2$, where T is the Time period of the leg as stated above. But, this value $J = m\Delta v_y$, as J is the change in vertical component of momentum. This impulse must have been received from the very beginning of walking/running. Hence, from here we can calculate the initial vertical velocity $v_y(0) = \Delta v_y/2 = J/2m = (g - N/m)T/4$, where N will be $\max(0, mg - mv_x^2/L)$, depending on the horizontal speed v_x . Note that this function $\max(u, v)$ means the maximum out of u and v .

Okay so, from these calculations, we move on to the important variable of Power. The average power required to move forward at velocity v_x can be estimated by assuming that one has to supply the energy mgh to raise the center of mass through a height h each half cycle, and that none of this energy is recovered.

The average power is thus $P = 2mgh / T$. The height h can be calculated from $h = v_y(0)T/8$, from which the power can be calculated:

$$P = \frac{mgv_y(0)}{4} = (g - \max(0, g - v_x^2/L)) mgT/4$$

Continuing with our assumptions: -

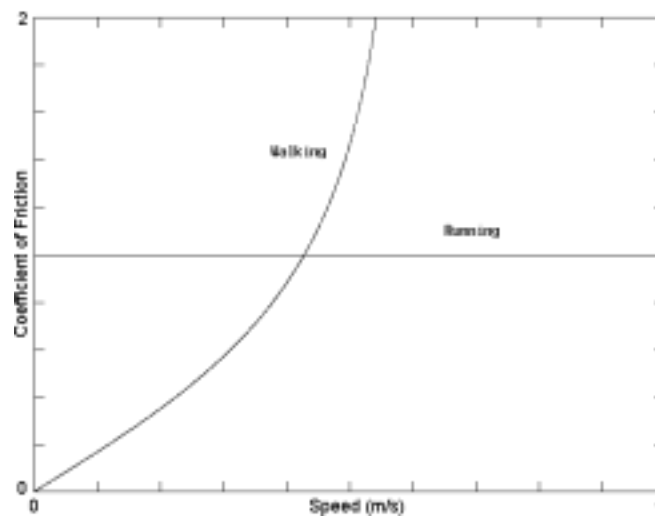
2A) During walking, the legs swing with their natural period, assumed to be given by $T = 2\pi [2L/3g]^{1/2}$, where $g = 9.8 \text{ ms}^{-2}$ (or 10 ms^{-2} , as per our convenience!) is the acceleration due to gravity, independent of walking speed, 'L' is the length of the leg. This assumes that the leg is a uniform rod, hinged at the top (a physical pendulum).

3A) During walking, energy is consumed in raising the centre of mass of the body once per step, and this energy is not recovered when the centre of mass is lowered again.

2B) During running, the assumptions are somewhat more complicated, as the legs do not oscillate anymore, but instead, the centre of mass moves in a parabolic trajectory, with maximum possible range, and if the surface of running does not suffice this condition, then



It is only defined as a transition of walking to running, or as falling on your nose with a thud! From here, the Power consumed during running is found to be $P = mgv_x/4$, which uses this fact of maximum range. Another variable in walking/running studies is the coefficient of friction of the surface. Look at the following graph, which studies walking/running speeds for a test-case human being, against the minimum coefficient of friction needed for that speed (Graph 1). Note that this is an Experimental Graph, so it may or may not agree with our model.



Graph 1

QUESTIONS:-

Q1. A man weighs 80 kg, and is 1.62 m tall. His legs are 1 m long. From the above model of walking/running, calculate his approximate minimum speed of running.

- A. Data is insufficient.
- B. 3.16 m/s
- C. 6.32 m/s
- D. 1.58 m/s



*Q2. Taking into consideration only the first assumption, Which of the following statements is/are true, with respect to our model of walking and running?

A. At very low walking speeds, it is reasonable to say that Power

required is $v_x^2 mgT/4L$

B. At running speeds, Power required is $mg^2T/4$

C. At the Transition speed, Power required is zero.

D. All of this calculation is dependent on air resistance, and we are wrong not to include it.

Q3. Taking all three assumptions into consideration, with $v_x = v$, which of the following is the correct expression of Power required for walking, P_w ?

A. $P_w = (mg / \pi)[3gL/2]^{1/2}\{1 - [1 - \pi^2 v^2/6gL]^{1/2}\}$

B. $P_w = (mg / \pi)[9gL/2]^{1/2}\{1 - [1 - \pi^2 v^2/6gL]^{1/2}\}$

C. $P_w = (mg / \pi)[3gL/2]^{1/2}\{1 - [1 - \pi^2 v^2/3gL]^{1/2}\}$

D. $P_w = (mg / \pi)[3gL/2]^{1/2}\{1 - [1 - \pi^2 v^2/6gL]^{1/2}\}$

*Q4. Which of the following statements is/are true, based on the above model?

A. At all horizontal speeds, walking consumes less power than running does.

B. After a particular horizontal speed, running consumes less power than walking.

C. After a particular horizontal speed, Walking consumes lesser power than running.

D. The speed at which both walking and running consume same power is

$v_c = (12/5\pi) [2gL/3]^{1/2}$



Q5. Statement 1: A person can transit from walking to running at a lower speed on the moon.

Statement 2: The velocity for transition is inversely proportional to square root of acceleration due to gravity

- A. Statement 1 is true, Statement 2 is true, and Statement 2 is the correct reason for Statement 1.
- B. Statement 1 is true, Statement 2 is true, but Statement 2 is not the correct reason for Statement 1.
- C. Statement 1 is false, Statement 2 is true.
- D. Statement 2 is false, Statement 1 is true.

Q6. Statement 1 : A shorter person can 'run' at lower speeds than a taller person

Statement 2 : There is a direct proportion between transition speed and square root of length of leg.

- A. Both Statements are True, Statement 2 is the correct reason for Statement 1.
- B. Statement 2 is false, Statement 1 is true.
- C. Statement 1 is false, Statement 2 is true.
- D. Both Statements are true, but Statement 2 is not the correct reason for Statement 1.

*Q7. From the graph in Graph 1, we can conclude that: -

- A. A coefficient of friction equal to 1 (at least approximately) in the graph is justified for our model of running.
- B. The coefficient of friction required in our model of running should decrease drastically with running speed.
- C. It is easier to walk on a very smooth surface than to run on it.
- D. None of the above can be concluded.



Q8. Assuming Optimum Running and Walking are according to our model, consider this story of a chase in the snowy flatland of Russia. Mr. Hare is standing on ice ($\mu=0.4$). 10 meters away from him, Mr. Tortoise is standing on Tarmac ($\mu=1$), and steals something. In which of the following scenarios, Tortoise can escape being caught by Hare?

- A. Hare starts running on ice, Tortoise starts running on tarmac.
- B. Hare starts walking on ice, Tortoise starts walking on tarmac.
- C. Hare starts walking on ice, Tortoise starts running on tarmac.
- D. In all of the above cases, Hare can catch Tortoise.



Neuralink

Passage

Neuralink as the name suggests creates a link between the Neurons inside our brain and a machine (smartphones or computers). It was started by Elon Musk in 2016 and it aims to “understand and treat brain disorders” and in the near future, will enable us to upload our brains to a server and live our lives digitally forever. It also talks about how we can use the internet without using our phone and download anything we want, be it your course textbooks or Video Games, directly into our brains!!

But how does he want to achieve this? Well, the idea is simple, he uses a couple of arrays of threads and electrodes which are implanted to the brain so as to create a scalable, high-bandwidth channel to transfer the signals from the brain to a computer. Also, the fact that our brain is a mere 12.4cm at its widest and 12 cm long, it's not that tough to actually get data from it using some electrodes.

These electrodes can then understand the electrical signal in your brain and translate them into an algorithm that a machine can read. This way Neuralink will be able to read what you are thinking and find a way for you to talk to machines without even opening your mouth. So no more calling out “OK Google” or “Alexa”.

But how do we power up such a small device? While current methods use USB power, we have made a hypothetical scenario of powering the N1 chip. It does it via the Seebeck effect, which is basically a phenomenon in which a temperature difference between two dissimilar electric conductors or semiconductors produces a voltage difference between those two substances.

Say the temperature is 33.5°C for the skin, 37°C for the brain, so the electrodes will be able to generate some electricity from this temperature difference to power up the microprocessor.
The relationship between the Voltage and Temperature difference is simple

$$V_s = ST$$



Here, V_s is the Voltage produced, T is the temperature difference and S is the Seebeck coefficient of the joint of the two materials.

Based on this, an important parameter known as the thermoelectric figure of merit or Z_t is derived via the formula-

$$Z_t = S^2 T / \rho \kappa$$

Symbols- electrical resistivity (ρ), and thermal conductivity (κ), absolute mean temperature (T)

QUESTIONS:-

Q1- Brain-Machine Interfaces basically communicate with the frontal cortex of the brain which is 2/5ths of the entire human brain. Consider that 126 cm^3 is the volume of the average brain. The 'threads' are also just 0.005 mm^2 in size and have a length of $20 \mu\text{m}$. How many electrodes will be required for our BMI, neglecting the folds in the brain? -

- a. 36.4lakh
- b. 45.8lakh
- c. 46.8lakh
- d. 52lakh
- e. 98lakh

Q2- We need to power up the chip and we are given that for the electrode compound, ρ is given as $2 \Omega\cdot\text{cm}$ and κ is $87.412 \text{ Wm}^{-1}\text{K}^{-1}$. Now say we need to power up Neuralink for 12 hours and consume 6 milliwatts of power and 2mA. What should be Z_t for the compound?

- a. 0.31
- b. 0.42
- c. 0.87
- d. 0.92
- e. None of these

*Q3- Why did we have to use such a conductive type of coating instead of simple electrodes?

- a. Reduce Impedance
- b. Reduce Inductance
- c. Increase Capacitance
- d. Dissipate excess charges

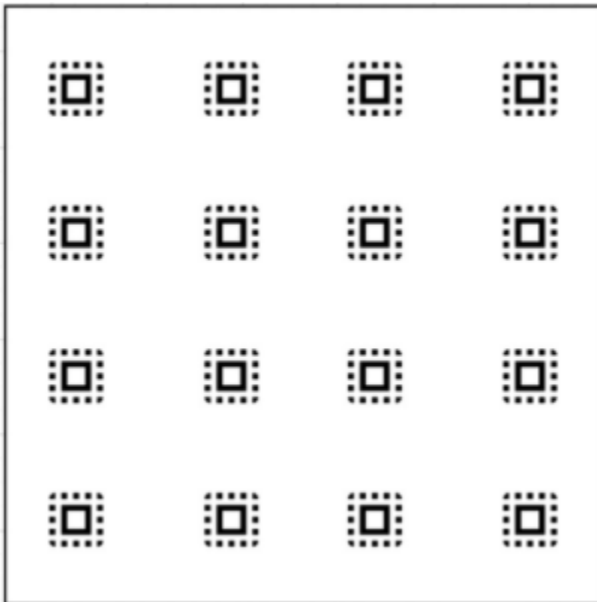
*Q4- The material used in modifying the properties of electrodes used in neuralink is a conductive organic material and the Z_t value is the same as you calculated in Q2. One of its constituents is a compound used as a medication for high blood potassium. Which of these statements about this constituent is true?

- a. Turns a red cabbage from purple to red in the presence of it.
- b. It cannot make hard water soft.
- c. The sodium fusion extract of the compound with sodium nitroprusside gives purple color.
- d. When mixed with it, gives white ppt with a barium chloride solution

*Q5- The electronics used in neuralink technology are mainly ASICs (a different type of IC) which are integrated onto the PCB using a certain technique which includes inverting the ICs and using adhesives to stick it on the PCB, while aligning their connection, such that it is electrically integrated. There is a certain property that gets affected because of the shortening of the wire length, and this leads to faster conduction of higher speed signals. Which property must have been varied to enable this?

- a. Decrease in conductance
- b. Increase in capacitance
- c. Decrease in inductance
- d. Increase in resistance
- e. None of these

Q6- In the above concept of inverting chips, the material of the IC and PCB is different, and therefore, thermal expansion must be taken care of to maintain the connections. The PCB is square of side 45 mm and IC is square of side 5 mm. The alpha (coefficient of thermal expansion) of the IC is 0.7 and for the PCB is 0.35. Consider both experiences the same temperature difference of 8 C when operated. As the ICs expand more than the PCB, there should be a safe gap between the ICs so that they do not make contact with each other even after expanding and even do not cross the borders of the PCB. What should be the gap between these ICs?



- a. More than 6.5 mm
- b. More than 6.75 mm
- c. More than 7 mm
- d. More than 6.25 mm
- e. None of these

Q7- We wanted to help a paralyzed boy after a terrible car crash to again play his beloved game Valorant, but he would have to move his fingers to control the game controller. But will he be able to win it against you, because it needs super fast reflexes and latency can be a major issue because the chip has to first get signals from your brain then transmit it directly to the game console via WiFi which has a 500Mbps uplink speed and a ping of 1ms and the game requires 2MB to be sent? Given that the data analysis rate (spike detection from neural impulses) is 1Gbps.

Say the reaction time to move a finger is 20ms, time to react in the game is 5ms, and 2.5 million neurons @10 signals per second are fired, also 1.2 meters is the distance between the finger and the brain. 50m/s is the speed of signal transfer by myelinated neurons while it's just 2m/s for unmyelinated neurons. The controller also has a latency of 11ms.

- a. You win
- b. The boy wins
- c. Both